

ABDELLAH ELAZZAOU

AI Engineer | Data Engineer | Machine Learning | Computer Vision | Big Data | LLM & AI Agents

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PROFESSIONAL SUMMARY

Driven AI & Data Engineering student at ENSA Agadir with end-to-end expertise across the full AI and data stack: building LLM-powered agents and RAG systems with LlamaIndex & LangGraph, deploying real-time computer vision with YOLO, and engineering scalable data pipelines with Apache Spark, Kafka, and Airflow.

WORK EXPERIENCE

- AI & Computer Vision Intern - Orange Summer Challenge 2025 Winner – Orange Morocco Jul – Sep 2025
 - Built a hybrid YOLO11 + CNN pipeline for tomato disease detection and classification
 - Managed large-scale image datasets: collection, annotation, preprocessing, augmentation, and train/val/test splitting at scale.
 - Executed the complete ML lifecycle from data engineering to model evaluation and stakeholder reporting.
- Data Science Intern – CodeAlpha (Remote) Jun 2025
 - Designed end-to-end data pipelines: ingestion, cleaning, transformation, EDA, and visualization using Pandas, and Seaborn.
 - Delivered analytical dashboards and reports communicating key business insights to stakeholders.

PROJECTS

- AI Personal Assistant – Gmail / Calendar / Sheets Manager
 - Multi-tool AI agent that autonomously manages Gmail, Google Calendar, and Google Sheets via natural language with human-in-the-loop.
 - Stack: LlamaIndex, Python, Google APIs, AI Agents, Agentic Workflows
- Health Assistant Chatbot – RAG System
 - Conversational health assistant using LangGraph and Retrieval-Augmented Generation, context-aware medical information.
 - Stack: LangGraph, RAG, Python, Vector Database, LLMs
- Construction Site PPE Safety Detection
 - Real-time CV system detecting missing personal protective equipment on workers and triggering safety violation alerts.
 - Stack: YOLOv11, OpenCV, Python, Object Detection
- Vehicle & License Plate Recognition + Dwell Timer
 - Surveillance system that detects vehicles, extracts plate numbers via OCR, and tracks time spent in front of a camera.
 - Stack: YOLO11, Pytesseract, OCR, Python, OpenCV
- Big Data Streaming Pipeline
 - Real-time pipeline using Kafka for event ingestion and Spark Streaming for large-scale processing, aggregation, and storage.
 - Stack: Apache Spark (SQL / Streaming / MLlib), Apache Kafka, Python, HDFS
- Gym Management System + AI Chatbot
 - Full-stack gym platform with member management, subscriptions, scheduling, and an LlamaIndex-powered chatbot for automated support.
 - Stack: Django, DRF, React.js, LlamaIndex, PostgreSQL

EDUCATION

- National School of Applied Sciences (ENSA), Agadir – Engineering – Data Science, Big Data & AI 2022 – 2027 (Expected)
- ALX Program – Data Engineering & AI Certification 2024 – 2025
- Baccalauréat – Sciences Physiques, Option Français – Mathematics, Physics & Analytical Sciences 2022

TECHNICAL SKILLS

- AI & ML: PyTorch, Scikit-Learn, CNN, YOLO ,Transfer Learning, Deep Learning, Model Fine-Tuning
- LLM & AI Agents: LlamaIndex, LangChain, LangGraph, RAG Systems, AI Agents, Prompt Engineering, Vector Databases
- Big Data & Streaming: Apache Spark (SQL / MLlib / Streaming), Apache Kafka, Apache Airflow, Docker
- Databases & ETL: PostgreSQL, MySQL, SQL, Data Modeling, Star Schema, ETL Pipeline Design, Data Warehousing
- Analytics & BI: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Excel, Chart.js
- Development: Python, Django / DRF, React.js, Git, REST APIs

CERTIFICATIONS

- Data Engineer Associate (DataCamp, Jan 2025) Associate Data Analyst in SQL (DataCamp, Feb 2025) Data Science with Python (DataCamp, Feb 2025)
- ALX Data Analytics (ALX, Sep 2025)

LANGUAGES

- Arabic – Native
- French – Intermediate
- English – Intermediate